352, HOLY CROSS ARTS AND SCIENCE COLLEGE FOR WOMEN

(Affiliated To Thiruvalluvar University)

DEPARTMENT OF COMPUTER APPLICATIONS

A Report on Project

Courier Management System (CMS)

A Report on Project submitted to the Department of Computer Applications

In Partial fulfillment of the requirements for the Degree of Bachelor of Computer Applications

Submitted by

Team ID : SWTID-2026-1164

Team Size 5

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Team member : Hajira D

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Courier Management System (CMS)

User Story:

Courier and logistics companies often struggle to efficiently manage shipments and deliveries due to reliance on manual processes and disconnected systems. The absence of a centralized shipment tracking mechanism makes it difficult to monitor parcel movement in real time, resulting in delayed delivery updates and poor visibility for customers. Delivery agent assignments are performed manually, increasing the chances of scheduling conflicts and operational delays. Furthermore, the lack of automated notifications prevents timely communication with customers, while the absence of performance analytics limits managerial insight into operational efficiency. Managing multiple branches becomes complex without an integrated platform, leading to coordination issues across locations. Collectively, these challenges result in customer dissatisfaction, reduced productivity, operational inefficiencies, and ultimately revenue loss for the organization.

1. INTRODUCTION

1.1 Project Overview:

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies.

The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

1.2 Purpose :

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnected	Stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	Uncertain

Problem Statements

PS-1

Branch Manager

I am
Branch Manager

I'm trying to
manage shipments efficiently

Because systems are disconnected

Which makes me feel
stressed

PS-2

Customer

I'm trying to
track parcels

But updates are
delayed

Because
no centralized system

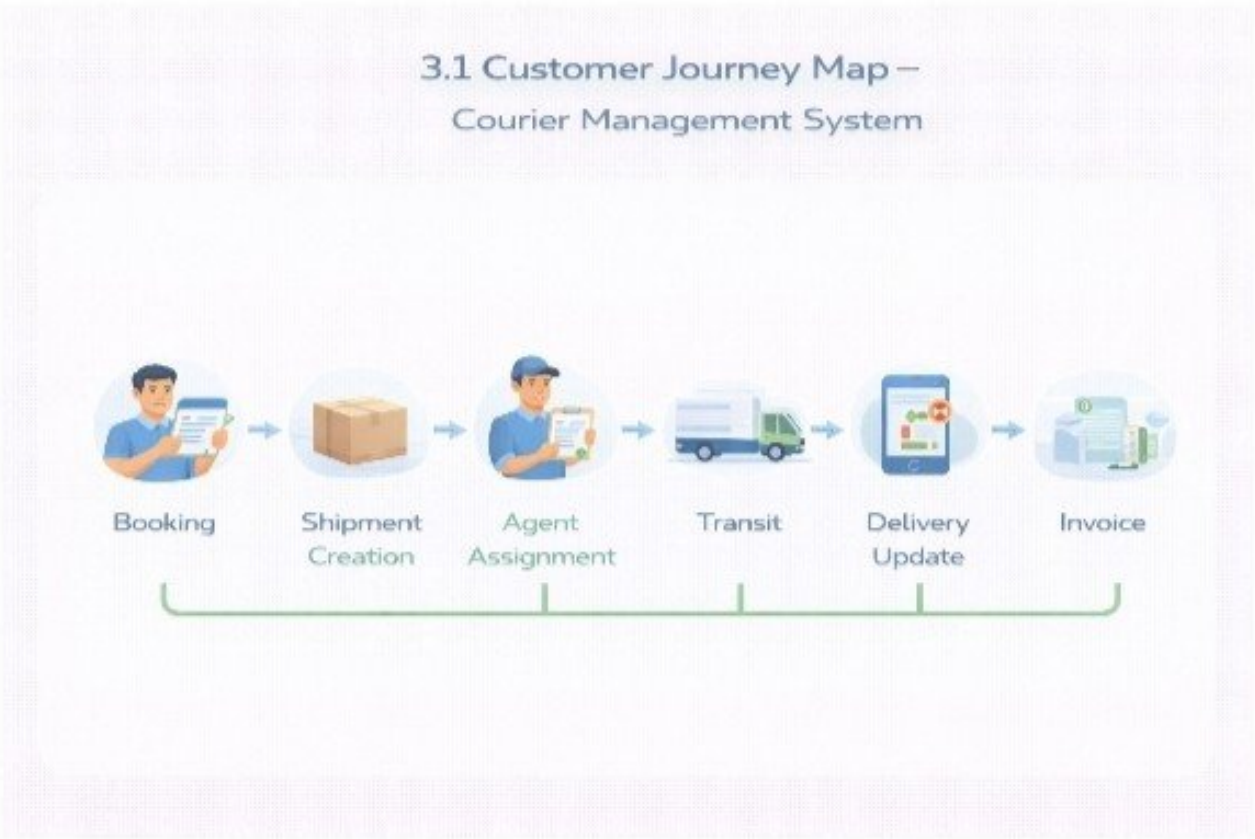
Which makes me feel
uncertain

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation

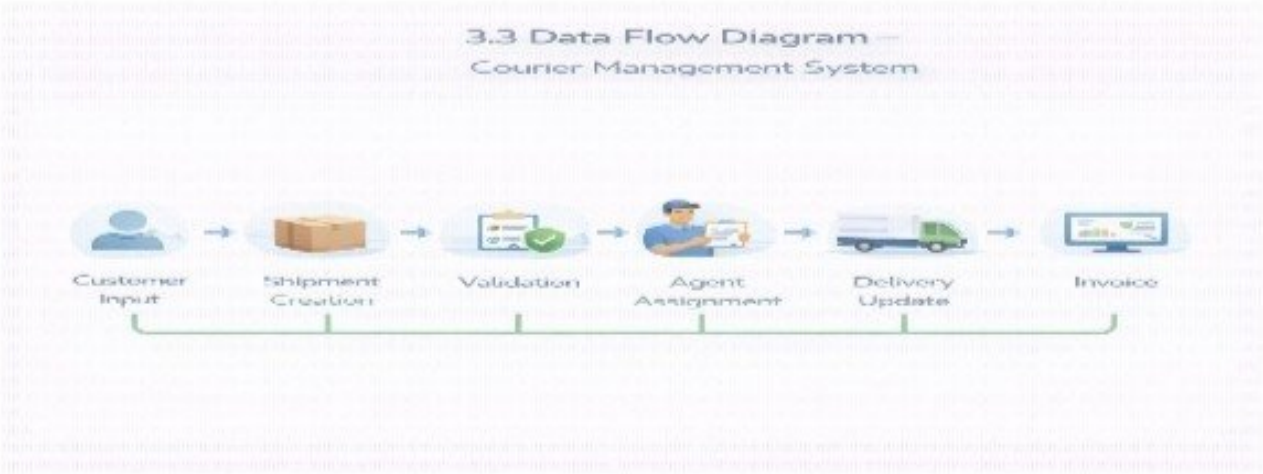
Non-Functional Requirements

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports



3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reprting	Reports & Dashboards

1. PROJECT DESIGN PHASE

1.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.

1.2 Proposed Solution

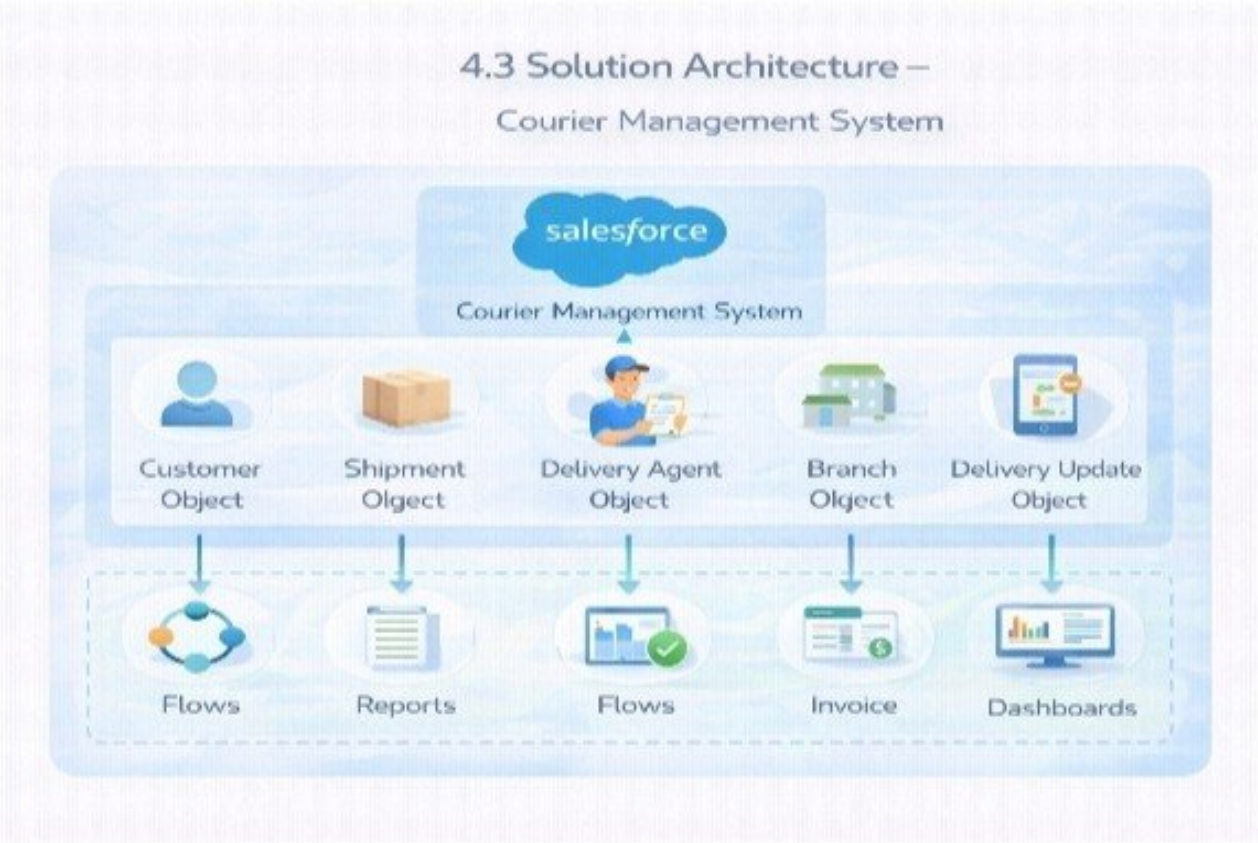
Functional Requirements

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking & billing
Impact	Faster deliveries
Business Model	Courier services
Scalability	Multi-branch ready

Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards



5.TEAM MEMBERS

Member1: HAJIRA D

Member2:JAYASRI N

Member3:PRABAVATHI M

Member4:BHUVANESWARI S

Member5:SWATHI M

2. PROJECT PLANNING & SCHEDULING

2.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

3.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Customer, Shipment, Delivery	5	High	Member2

4.

			Agent, Branch, and Invoice objects with proper relationships so that courier data is stored and managed efficiently.			
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Member5
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member1

Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	Medium	Member3

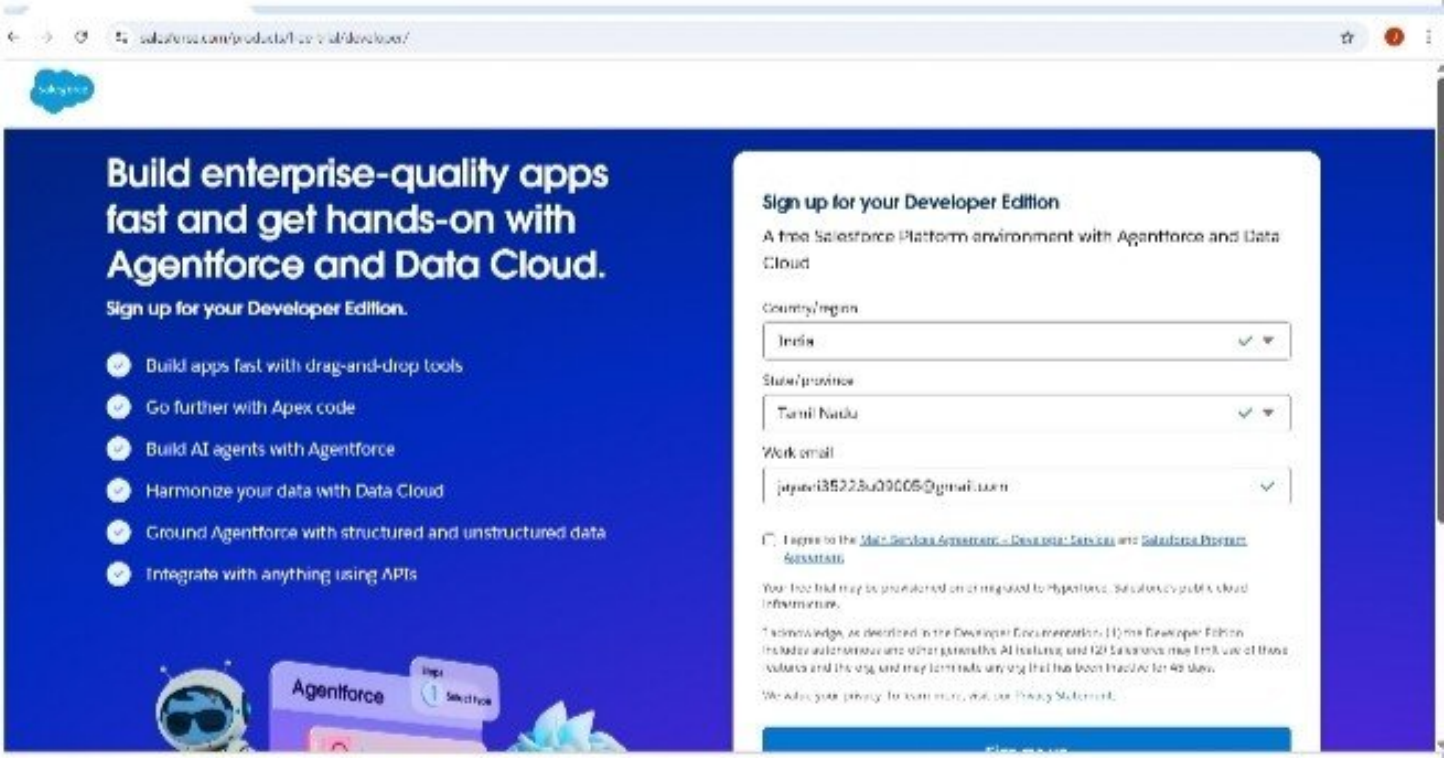
Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	02 March 2026	03 March 2026	2	07 March 2026
Sprint-2	20	6 Days	03 March 2026	04 March 2026		
Sprint-3	20	6 Days	05 March 2026	06 March 2026		
Sprint-4	20	6 Days	06 March 2026	07 March 2026		

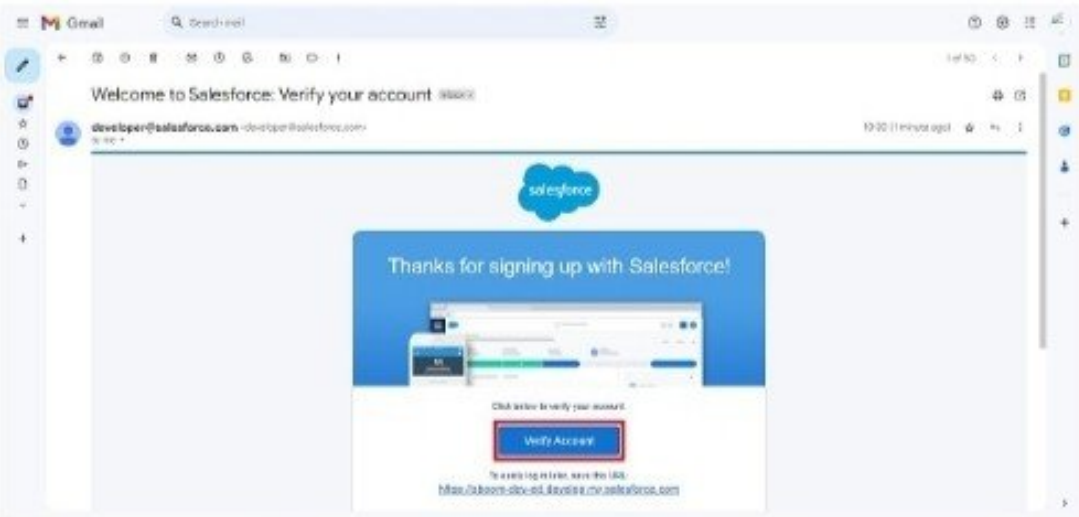
Milestone 1: Salesforce Account

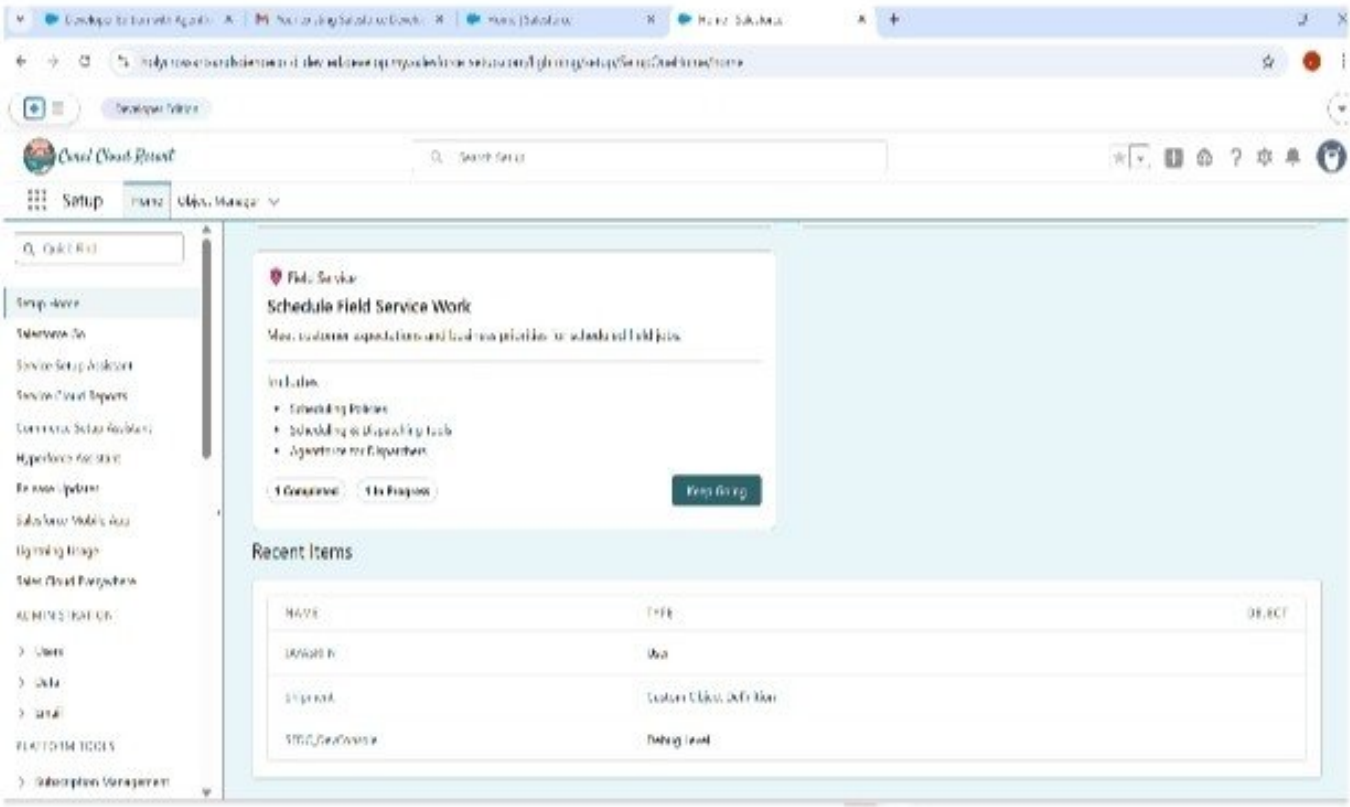
What Is Salesforce ?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.



Activity 2: Account Activation

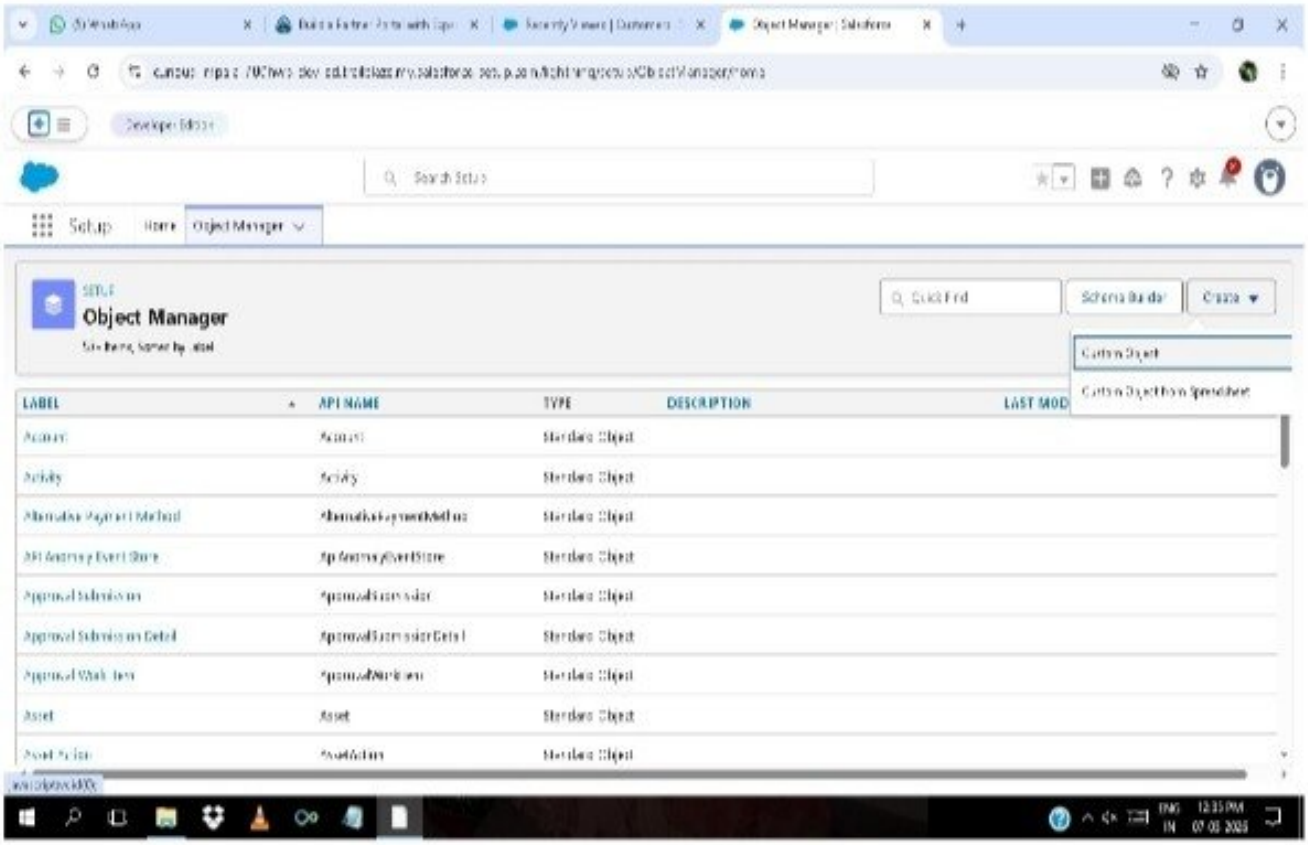




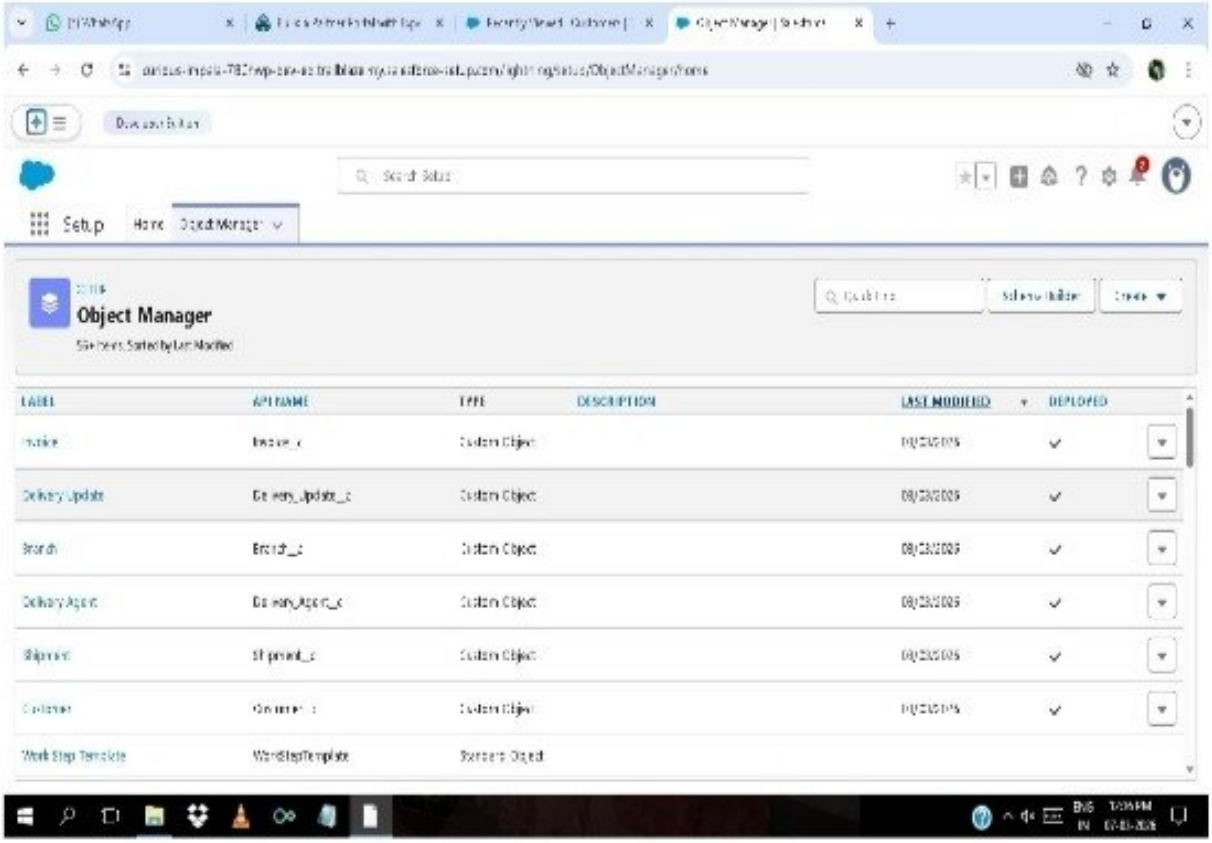
Milestone 2- Object:

Steps to Implement :

Activity 1: Create Customer Object



Activity 2: Create Shipment Object, Customer Object, Delivery_Agent Object, Branch Object, Delivery_Update Object, Invoice Object



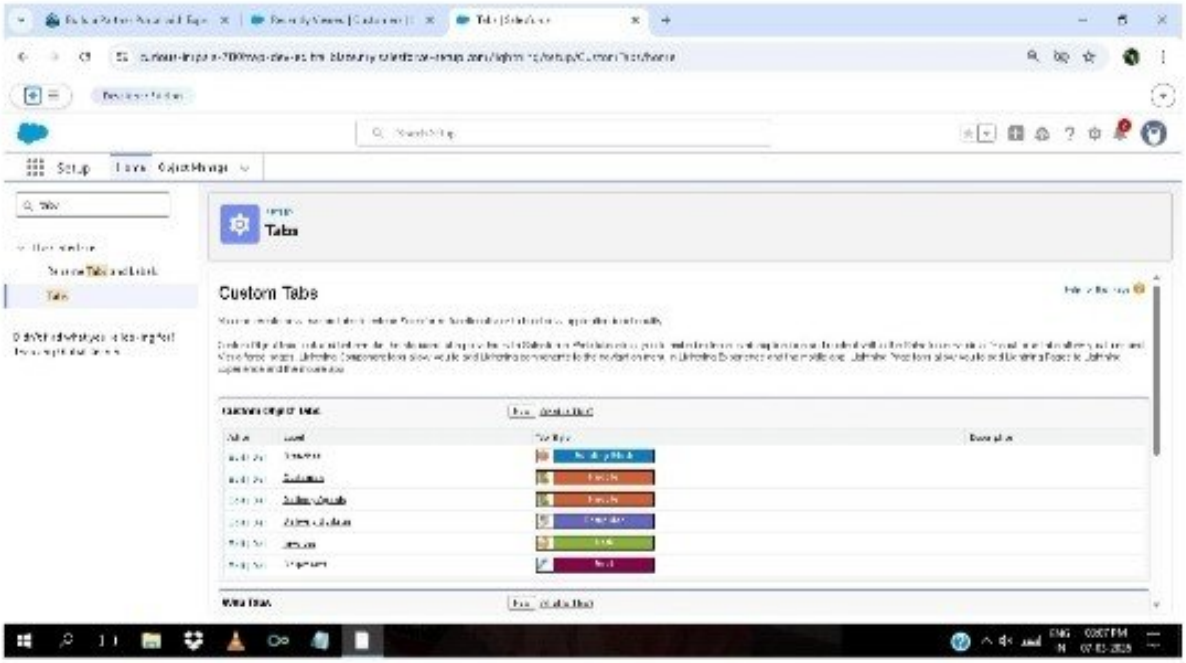
Milestone 3 - Tabs

Tab:

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1: Creating a Tab for Customer Object

Activity 2: Creating a Tab for Shipment Object, Delivery Agent Object, Branch object, Delivery Update object, Invoice Object Tabs

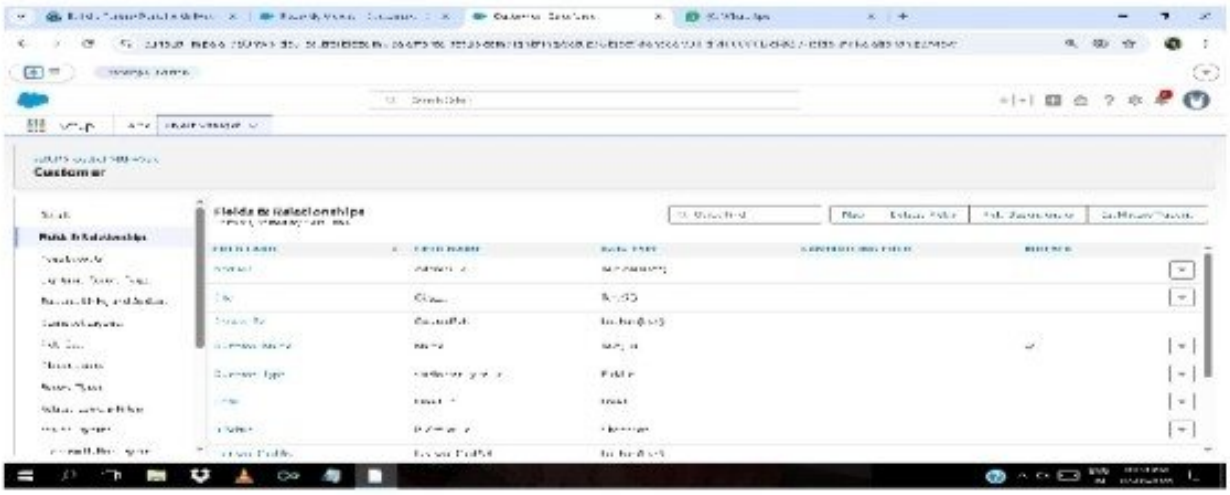


Milestone 4: Fields & Relationships

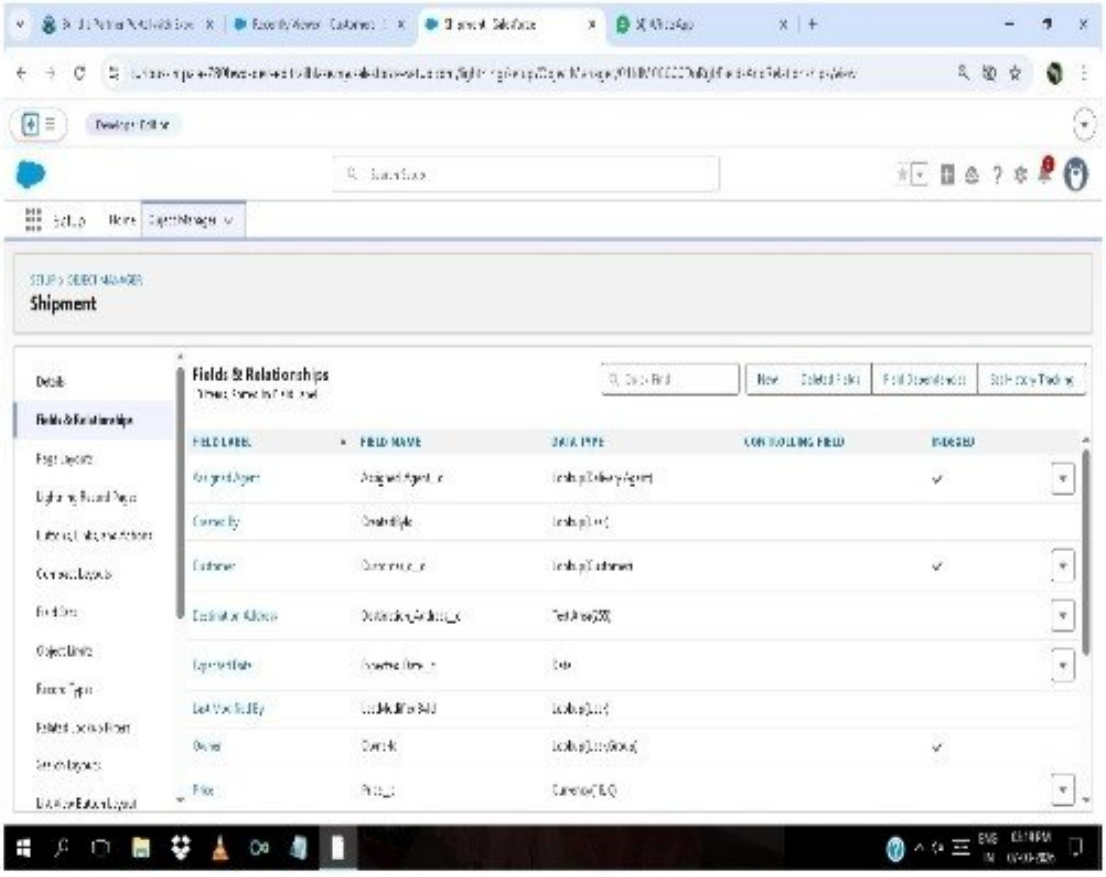
What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

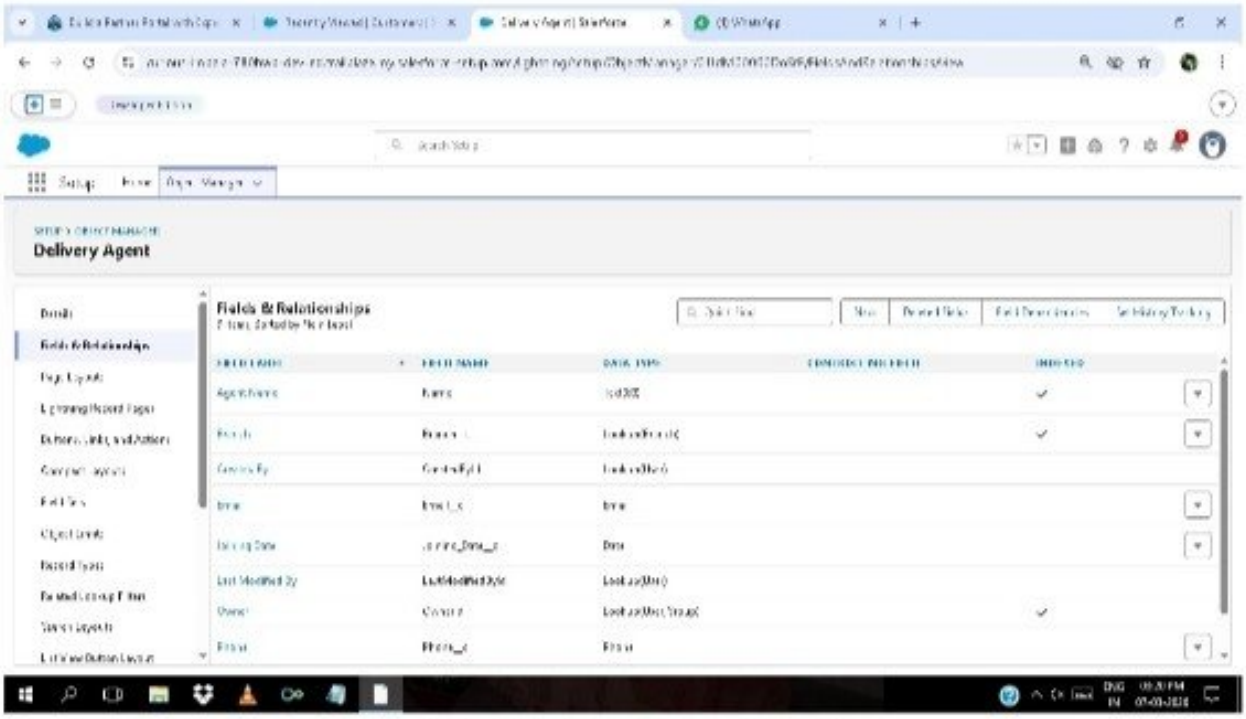
Activity 1: Creation of fields for the Customer Object



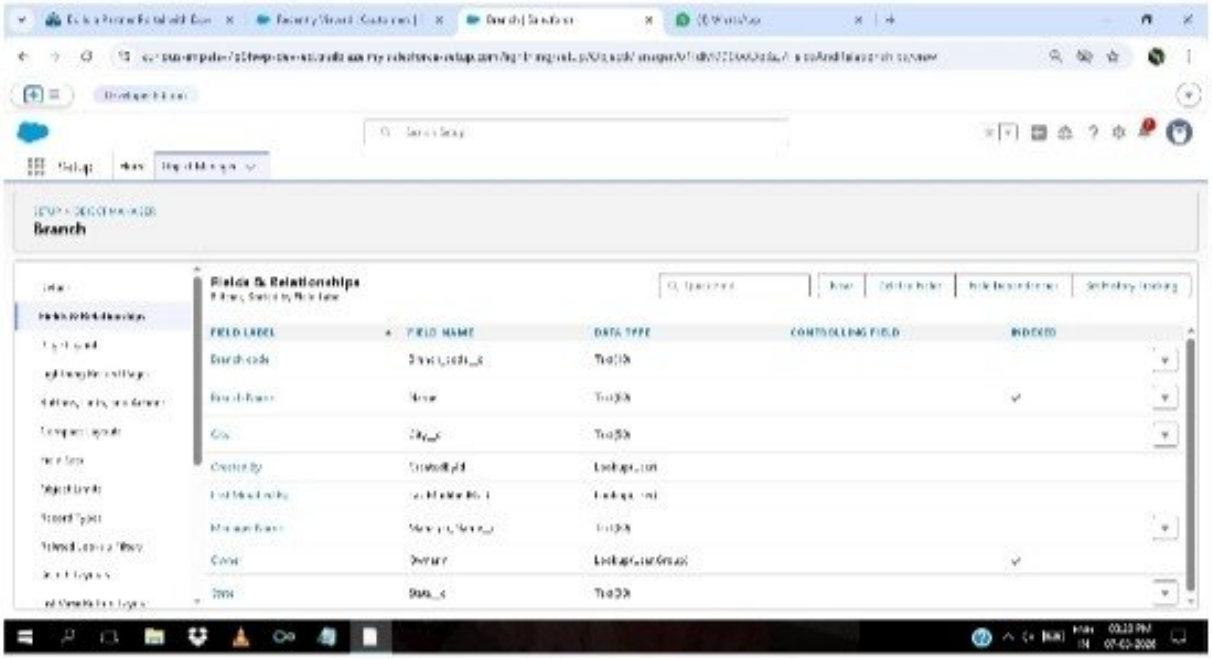
Activity 2:Creation of fields for the Shipment Object



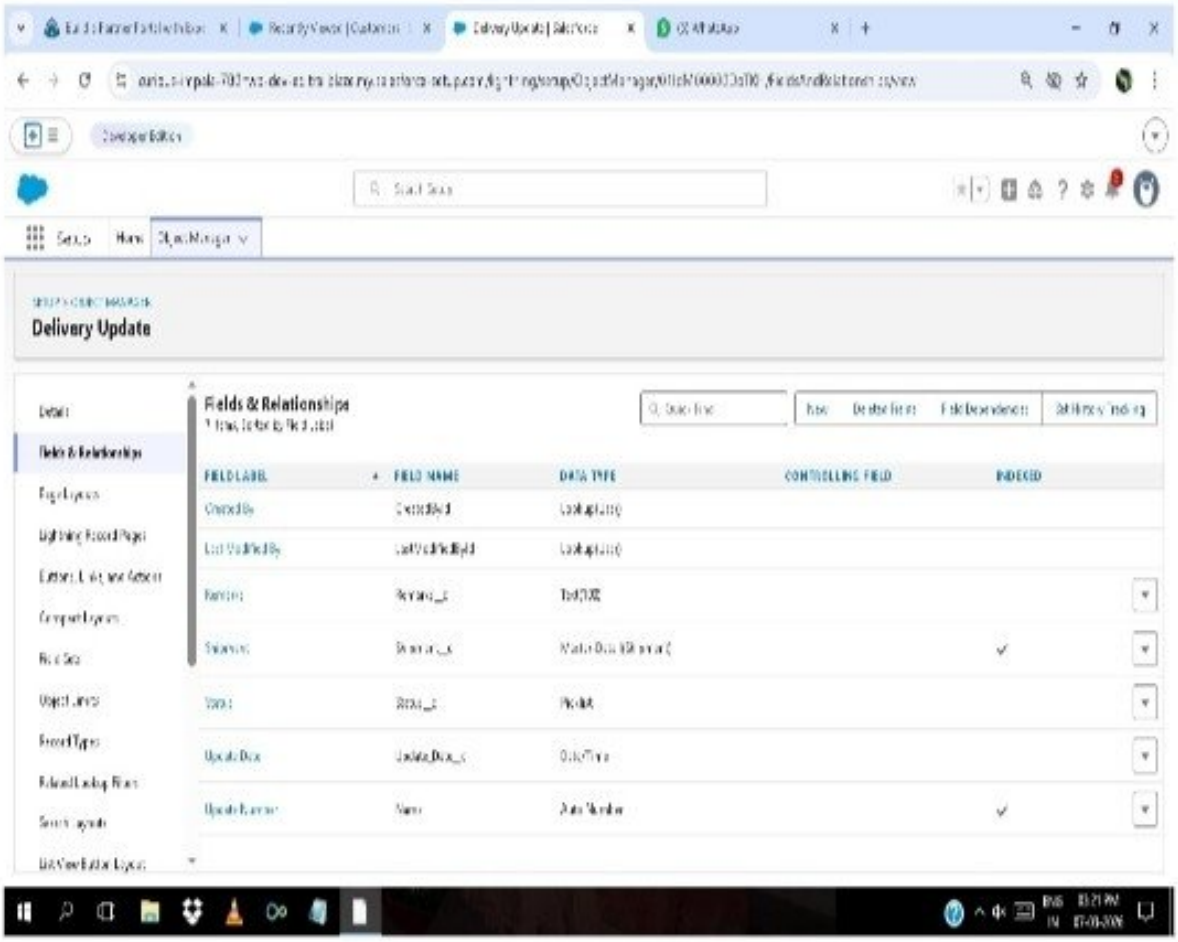
Activity 3: Creation of fields for Delivery_Agent Object



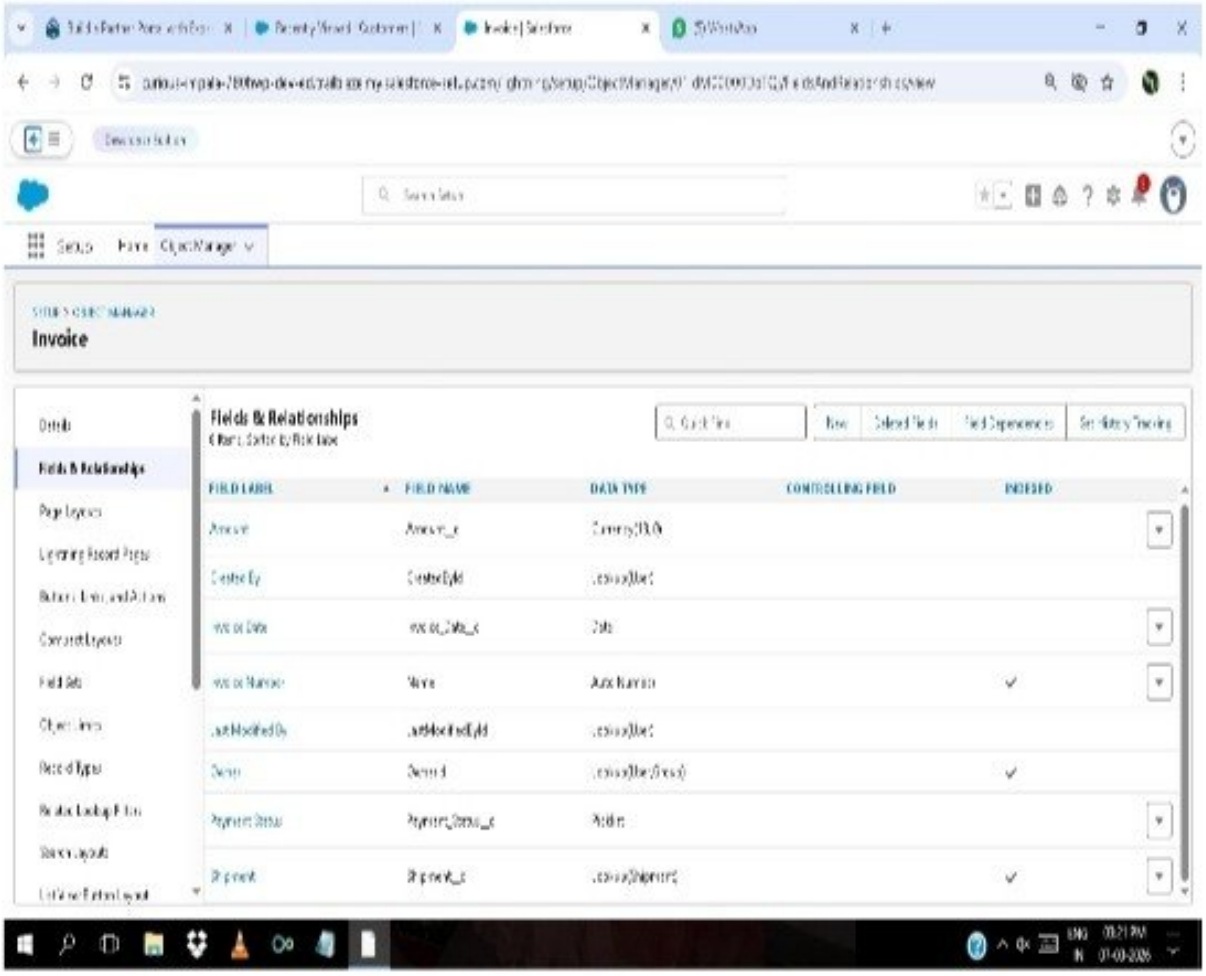
Activity 4: Creation of fields for Branch Object



Activity 5: Creation of fields for Delivery_Update Object



Activity 6: Creation of fields for Invoice Object



Relationships :

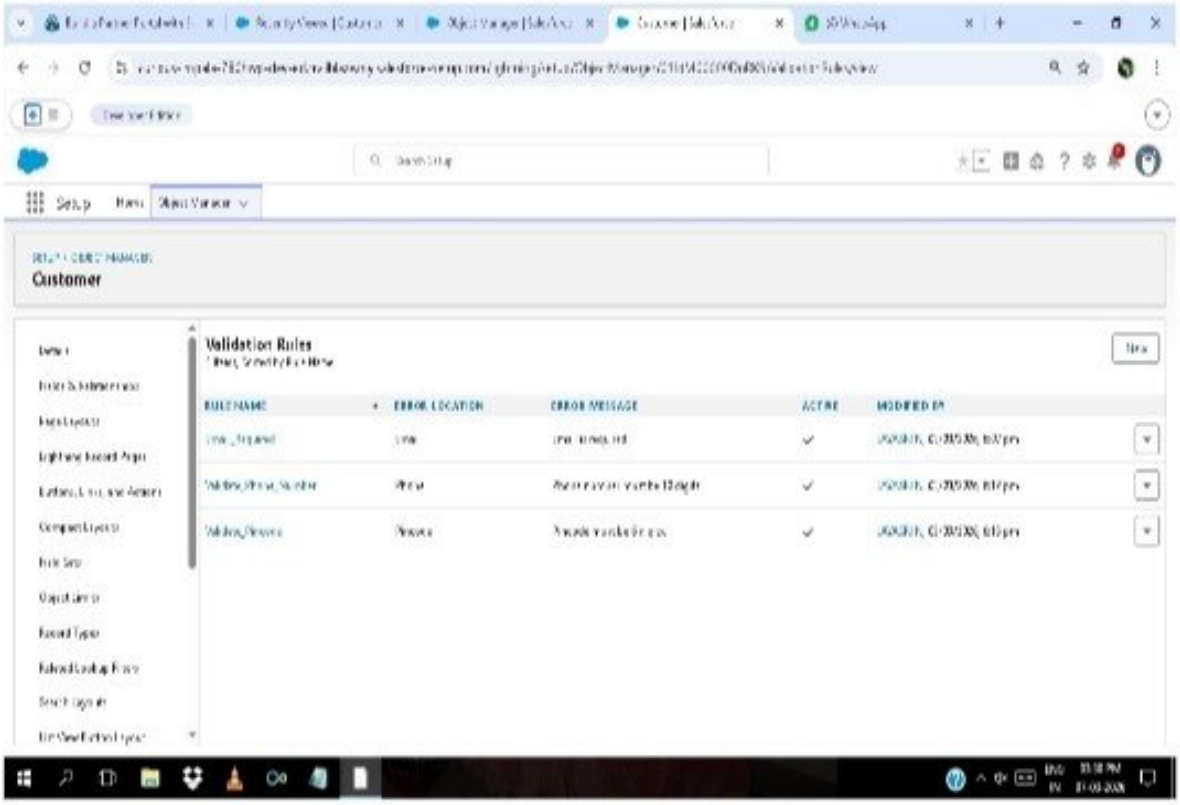
Child Object	Field	Parent Object	Type
Shipment	Customer_c	Customer	Lookup
Shipment	Assigned_Agent_c	Delivery_Agent	Lookup
Delivery_Agent	Branch_c	Branch	Lookup
Delivery_Update	Shipment_c	Shipment	Master-Detail
Invoice	Shipment_c	Shipment	Lookup

Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Activity 1: Create Validation Rules on Customer_c

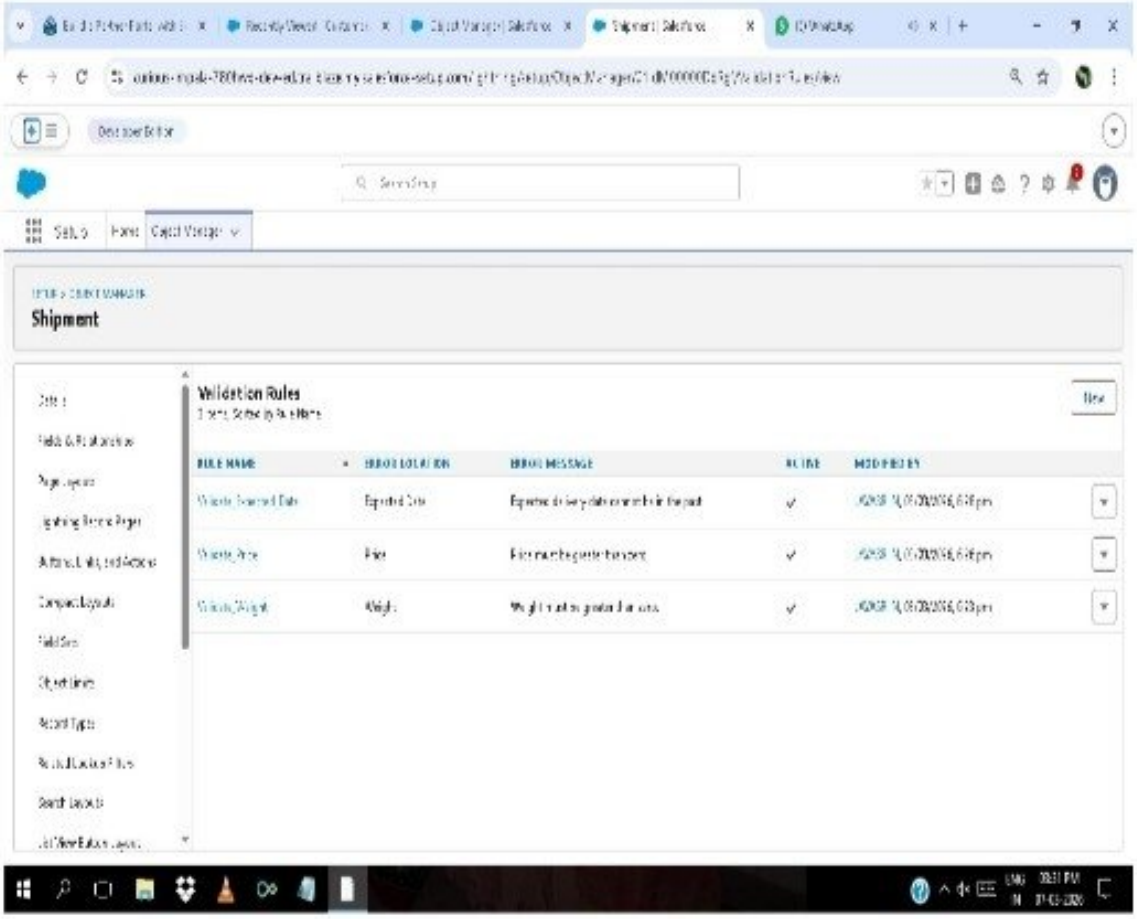
Validation Rules Summary



1.Customer

- Email Must Not Be Blank: `ISBLANK(Email_c)`
- Error: Email is required.
- Phone Must Be 10 Digits: `LEN(Phone_c) <> 10`
- Error: Phone number must be 10 digits.
- Pincode Must Be 6 Digits: `LEN(Pincode_c) <> 6`
- Error: Pincode must be 6 digits.

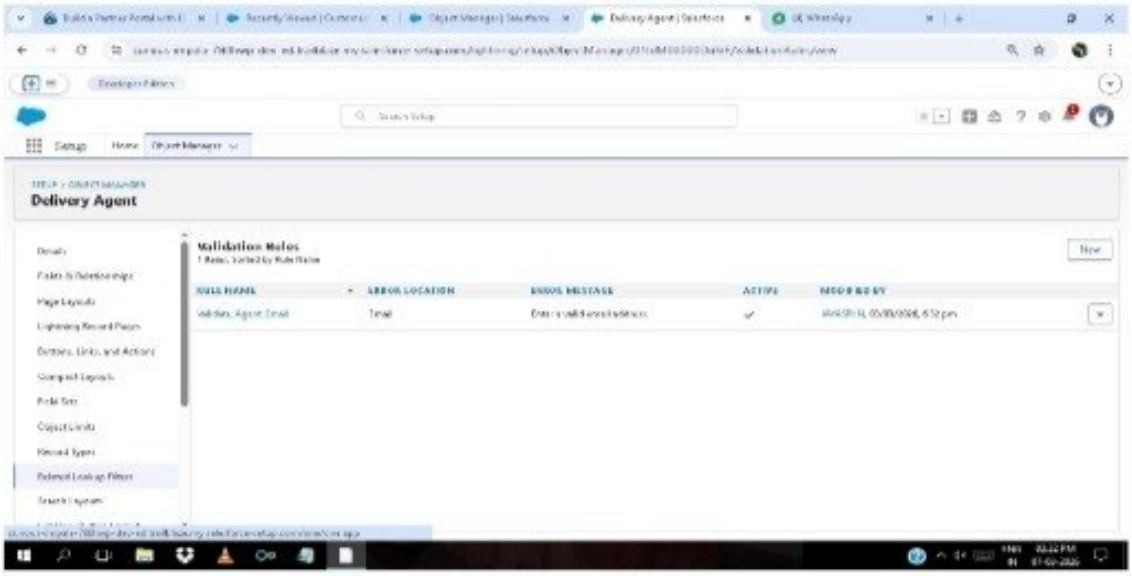
Activity 2: Create Validation Rules on Shipment____c



Shipment

- Weight Must Be Greater Than Zero: `Weight_c <= 0`
- Error: Weight must be greater than zero.
- Price Must Be Greater Than Zero: `Price_c <= 0`
- Error: Price must be greater than zero.
- Expected Date Cannot Be in the Past: `Expected_Date_c < TODAY()`
- Error: Expected delivery date cannot be in the past.

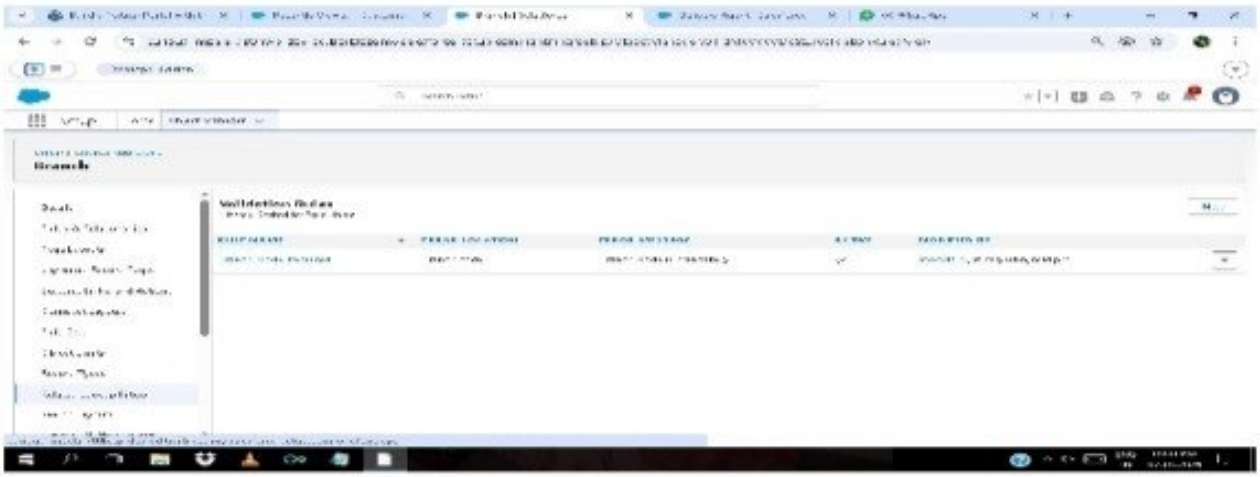
Activity 3: Create Validation Rules on Delivery_Agent_____c



3. Delivery_agent
- Email Must Be Valid: `NOT(CONTAINS(Email_c, "@"))`
- Error: Enter a valid email address.

Activity 4: Create Validation Rules on Branch_____c

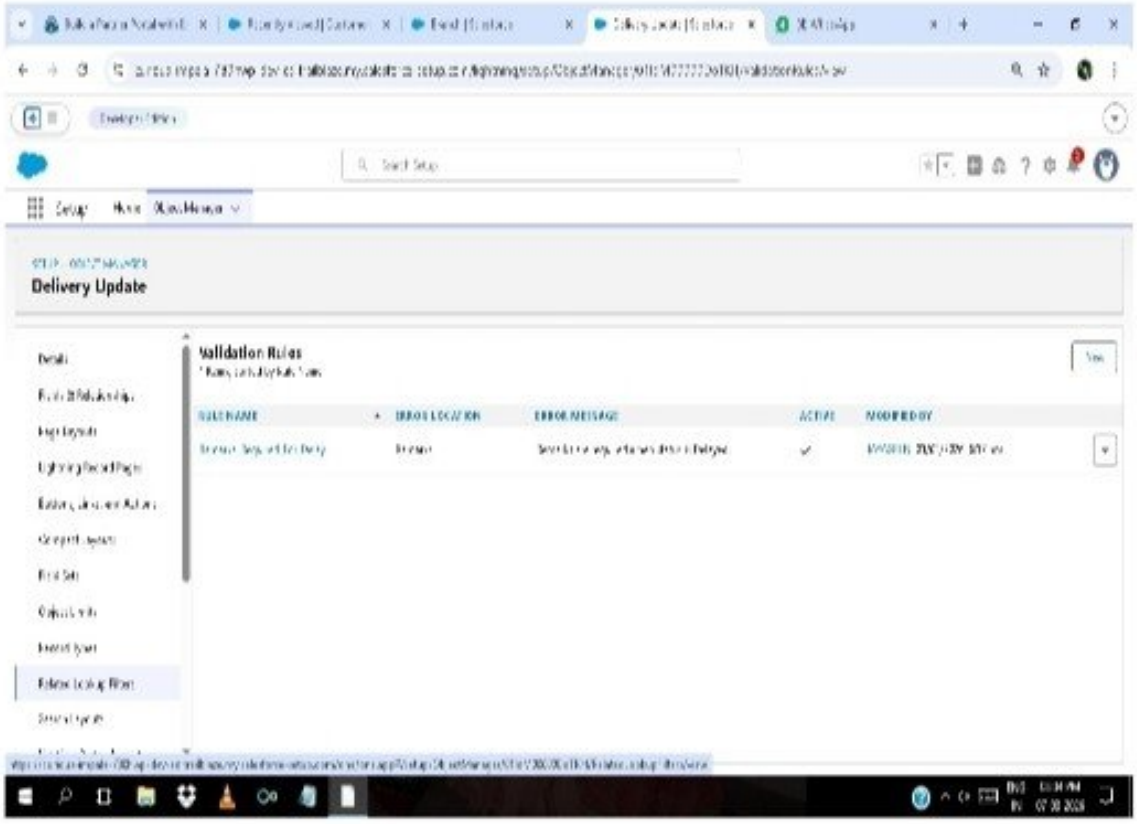
4. Branch
- Branch Code Must Not Be Blank: `ISBLANK(Branch_Code_c)`
- Error: Branch Code is mandatory.



Activity 5: Create Validation Rules on Delivery_Update_____c

5. Delivery_Update

- Remarks Required If Status is Delayed: `AND(ISPICKVAL(Status_c, "Delayed"), ISBLANK(Remarks_c))`
- Error: Remarks are required when status is Delayed.



Milestone 6 - Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.

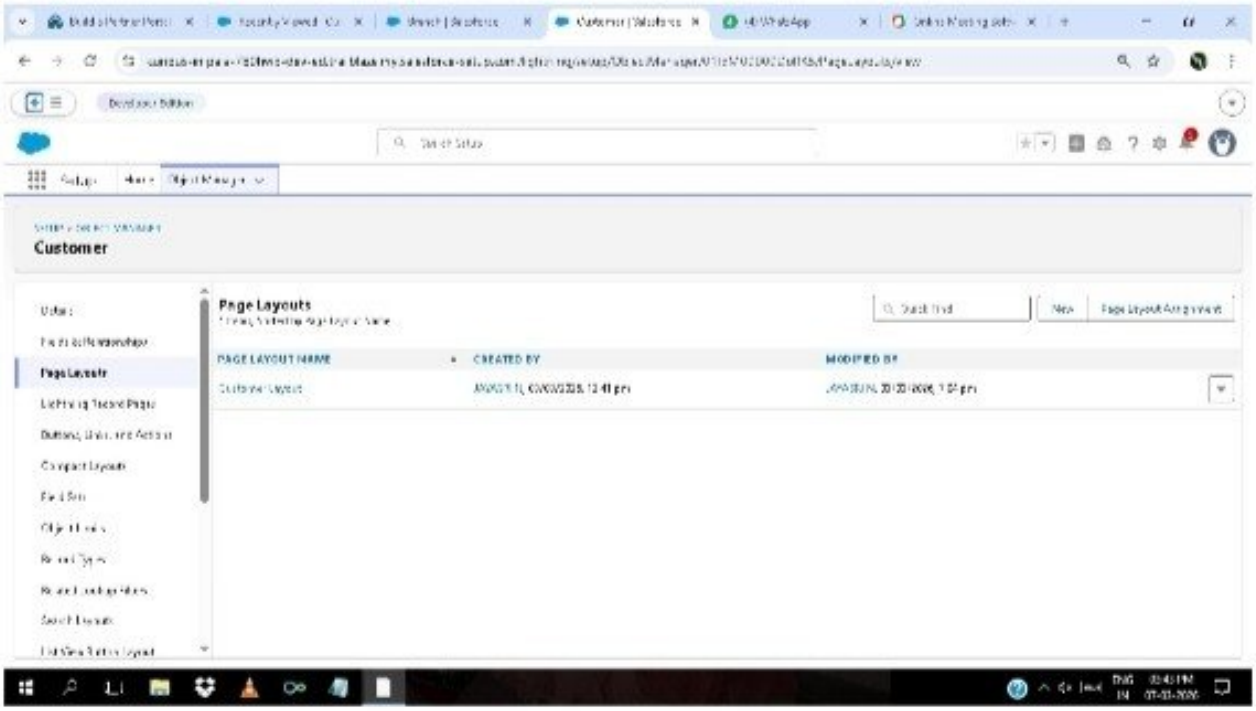
They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.

By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

Activity 1: Edit Page Layout for Customer____c

Page Layout Configurations

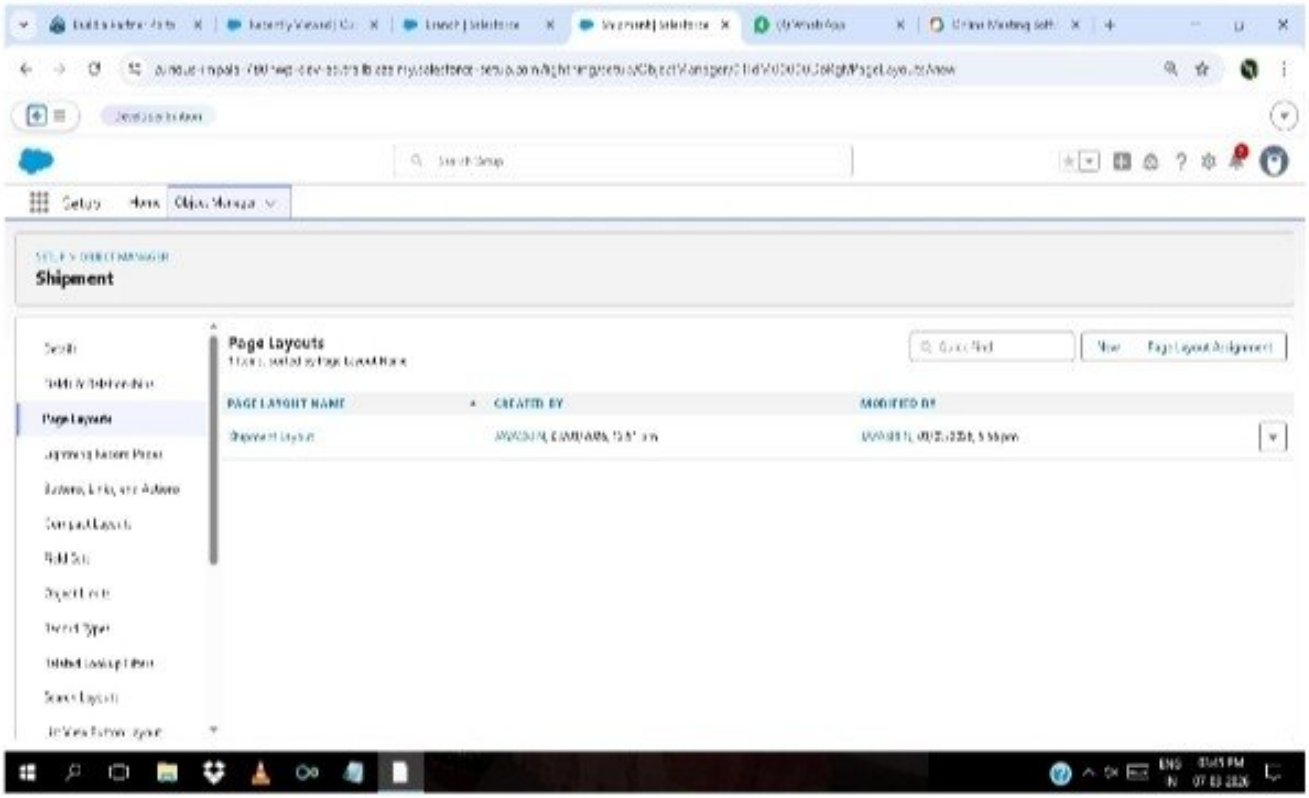
1. Customer
   - Sections:
   - Customer Information: Name, Type, Email, Phone, Is Active
   - Address Details: Address, City, State, Pincode



Activity 2: Edit Page Layout for Shipment __c

2. Shipment

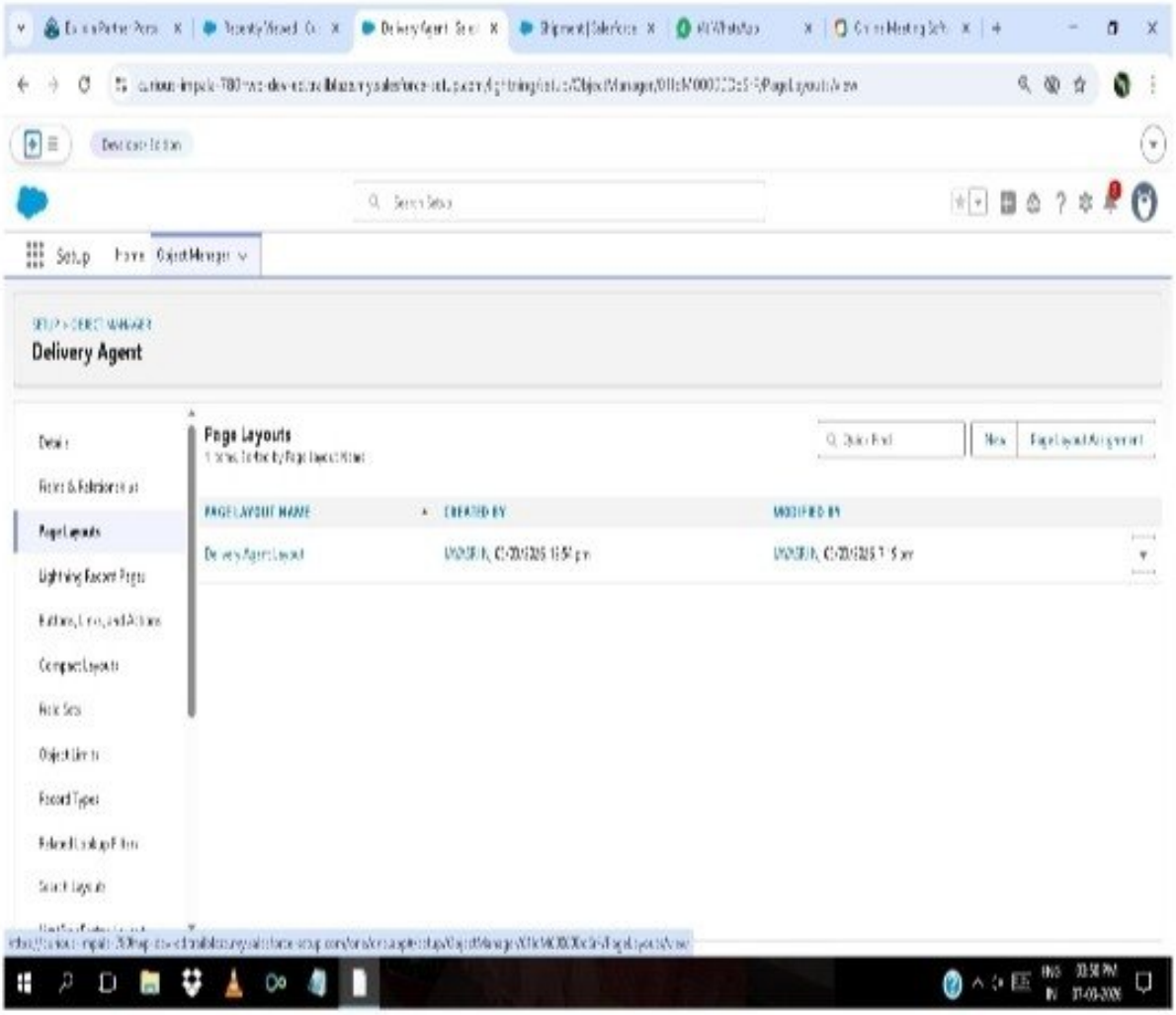
- Sections:
- Shipment Details: Tracking Number, Customer, Type, Weight, Price, Status
- Address Information: Source Address, Destination Address
- Delivery Information: Expected Date, Assigned Agent



Activity 3: Edit Page Layout for Delivery_Agent_____c

3. Delivery_Agent

- Sections:
- Agent Details: Name, Email, Phone, Branch, Joining Date
- Related Lists: Shipments, Delivery Updates



Activity 4: Edit Page Layout for Branch_____c

Steps:

- Setup → Object Manager → Branch_c
- Page Layouts
- Open default Branch Layout

Sections:

Branch Information

- Branch Name
- Branch Code
- City
- Stat

Activity 5: Edit Page Layout for Delivery_Update_____c

Steps:

- Setup → Object Manager → Delivery_Update_c
- Page Layouts
- Open default Delivery Update Layout

Milestone 7 - Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

Scenario:

Whenever a Delivery_Update_c record is created, automatically update the related Shipment_c Status with the latest status from Delivery Update.

Auto-Update Shipment Status Flow

1. Start:

- Object: Delivery_Update__c
- Trigger: Record created
- Conditions: None (always run)
- Optimize for: Actions and Related Records

2. Update Shipment:

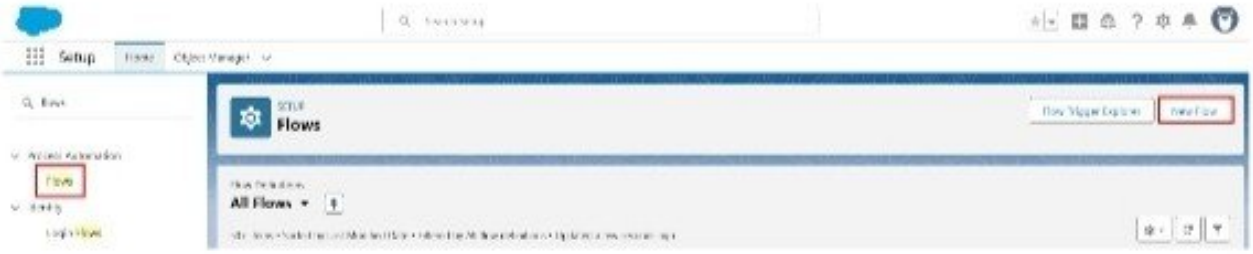
- Element: Update Records
- Label: Update_Shipment_Status
- Object: Shipment__c
- Filter: Id = {!$Record.Shipment__c}
- Update: Status__c = {!$Record.Status__c}

3. Save & Activate:

- Save as: Auto_Update_Shipment_Status
- Activate the flow.

Step 1: Open Flow Builder

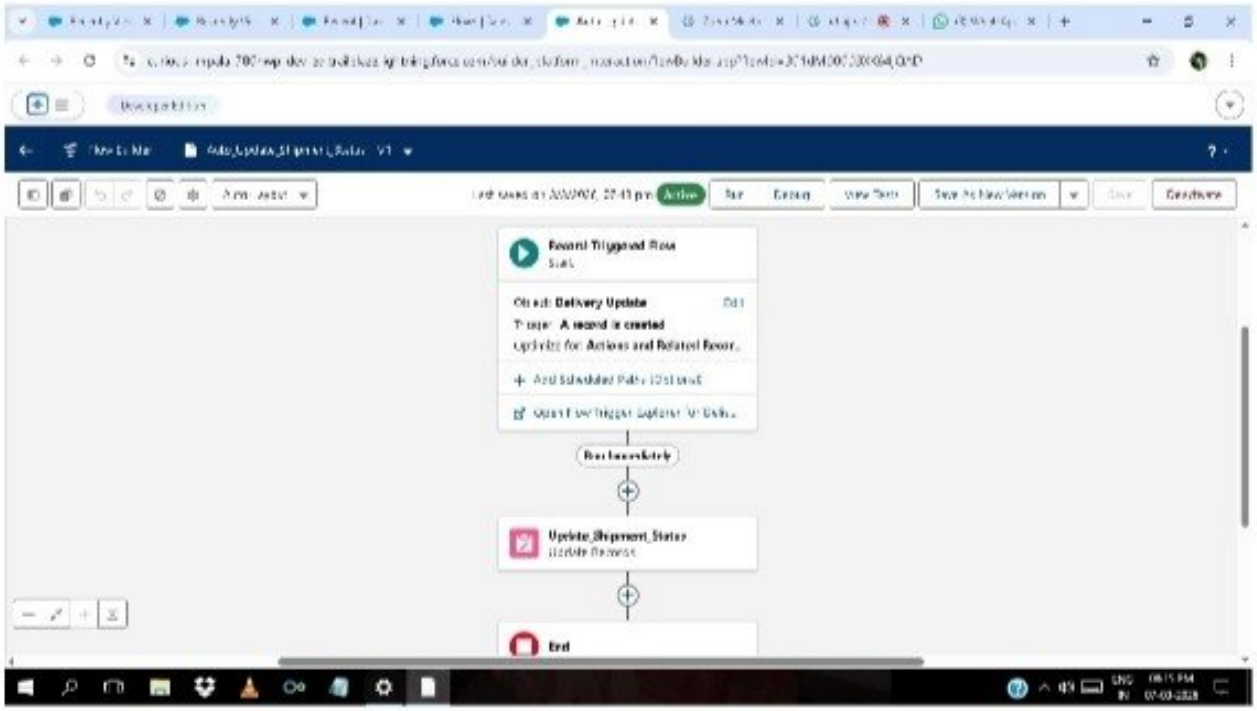
- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows
- Click the New Flow button



- Select Record-Triggered Flow
- Click Create

Shipment Field	Value Source
Status_c	{!$Record.Status_c}

Final flow report



Milestone 8 - The Lightning App :

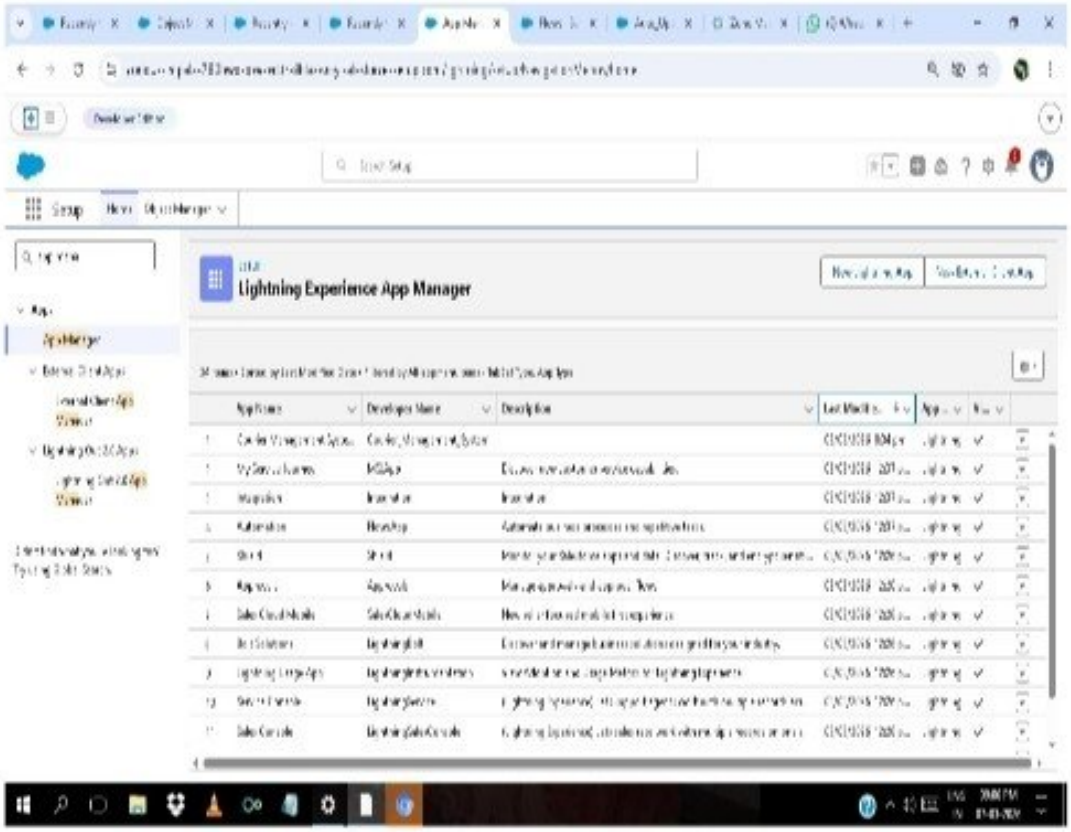
What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

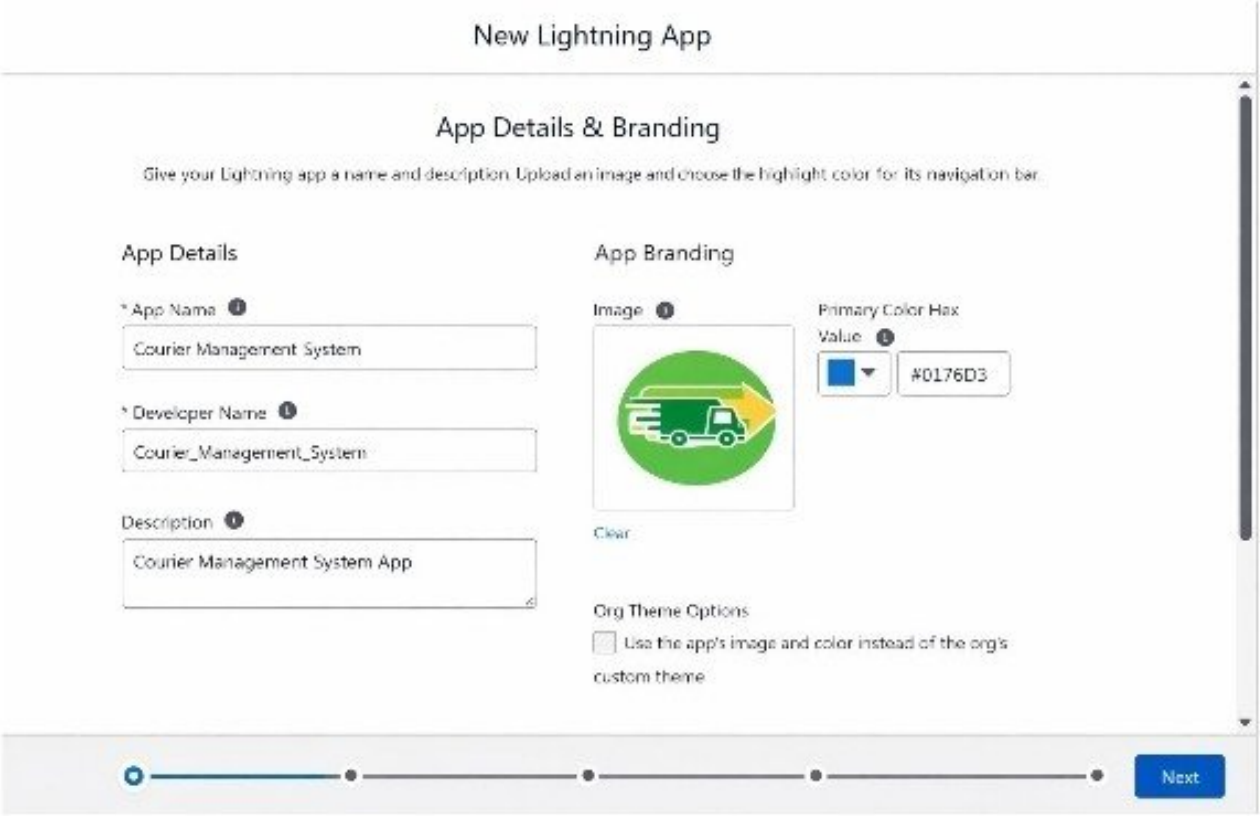
Activity 1: Create a Lightning App

Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.



2. Click New Lightning App.
    - Enter the Courier Management System as the App Name.
3. Upload an image or icon for the app, then click Next.



4. Under App Options, leave the default selections and click Next.
5. Under Utility Items, leave as it is and click Next.
6. From Available Items, select the following and move them to Selected Items:
    - Customer
    - Shipment
    - Delivery Agent
    - Branch
    - Delivery Update

- Invoice
- Reports
- Dashboards

7. Click Next.
8. From Available Profiles, select System Administrator and move it to Selected Profiles.
9. Click Save & Finish.

Milestone 9- Profile:

What is a profile?

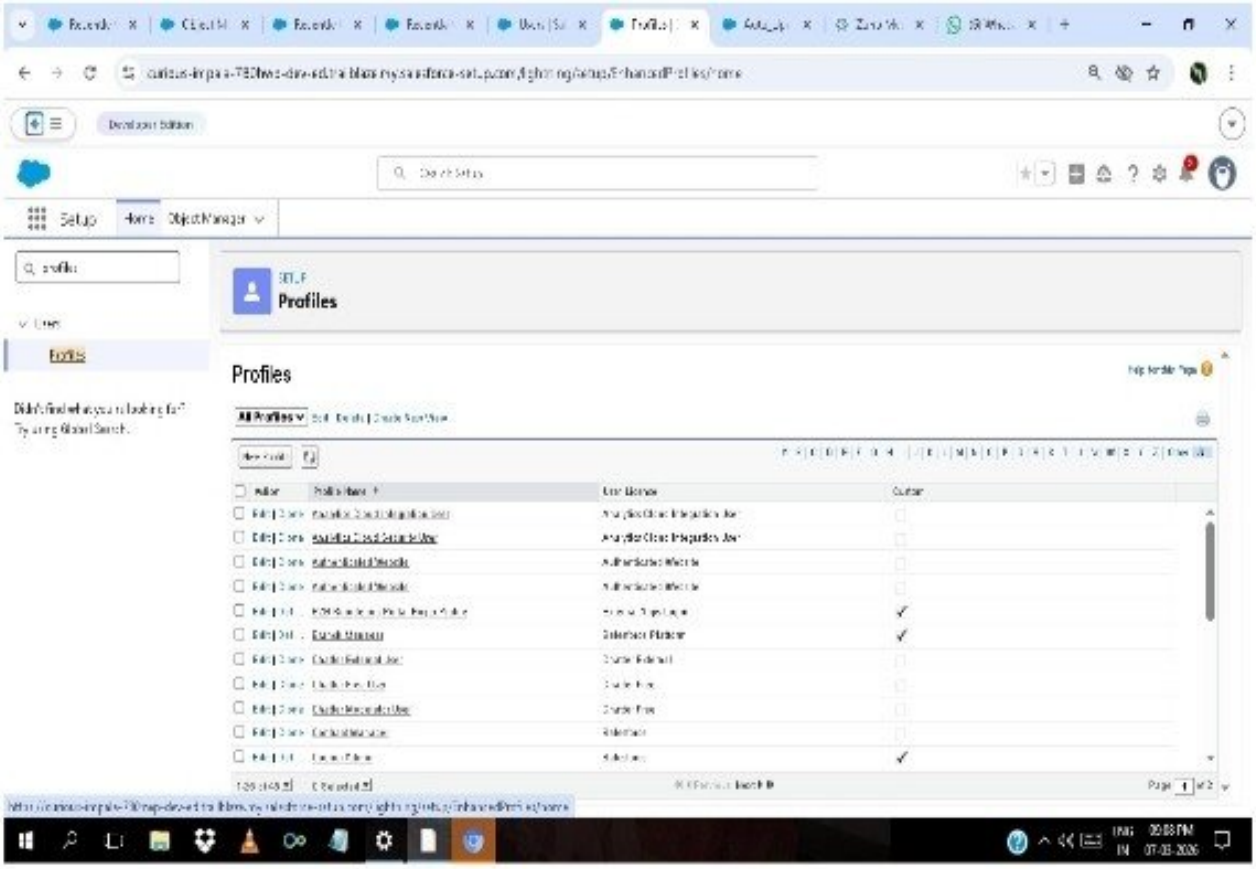
A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but users can be assigned a single profile at a time.

Activity 1: Create a Custom Profile – Courier Admin

Create a Custom Profile:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.



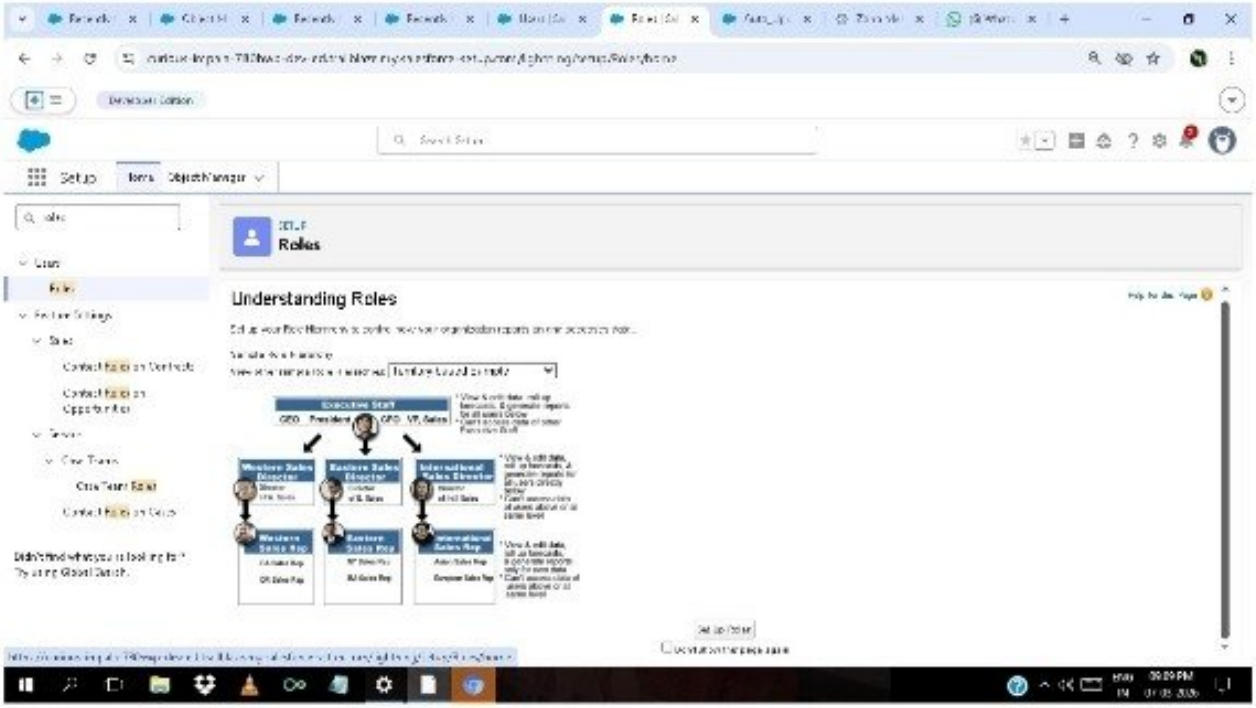
Milestone 10- Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system

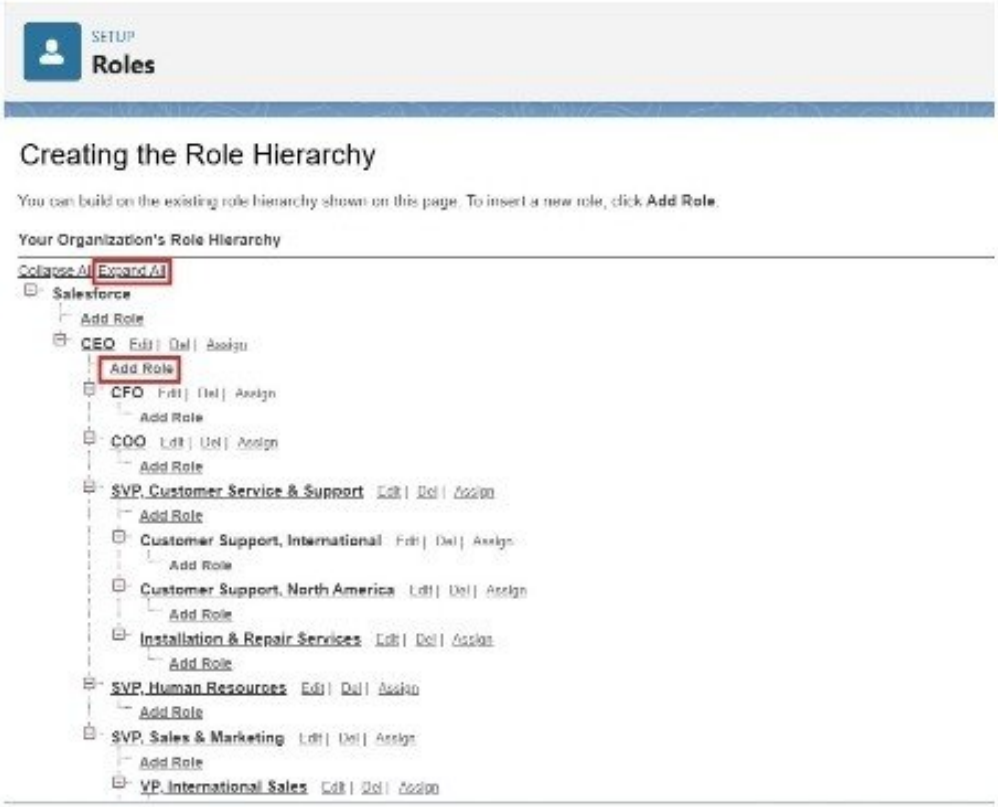
Activity 1: Creation of Courier Admin Role

Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.
2. Select Set Up Roles.



3. Click on Expand All and click Add Role under CEO.



Activity 2: Creating Branch Manager Role Under Courier Admin Role

Create another role:

1. Create a role with the name “Branch Manager”.
2. Below the Courier Admin, click on Add Role.
3. Enter the label name as “Branch Manager” and the Role Name will automatically populate. Click Save.

Activity 3: Creating Delivery Agent Role Under Branch Manager Role

Steps:

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.
    - Role Name will auto-populate.
3. Click Save.

Activity 4: Creating Customer Support Role Under Branch Manager

Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. Click Save.

Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

Milestone 11- User :

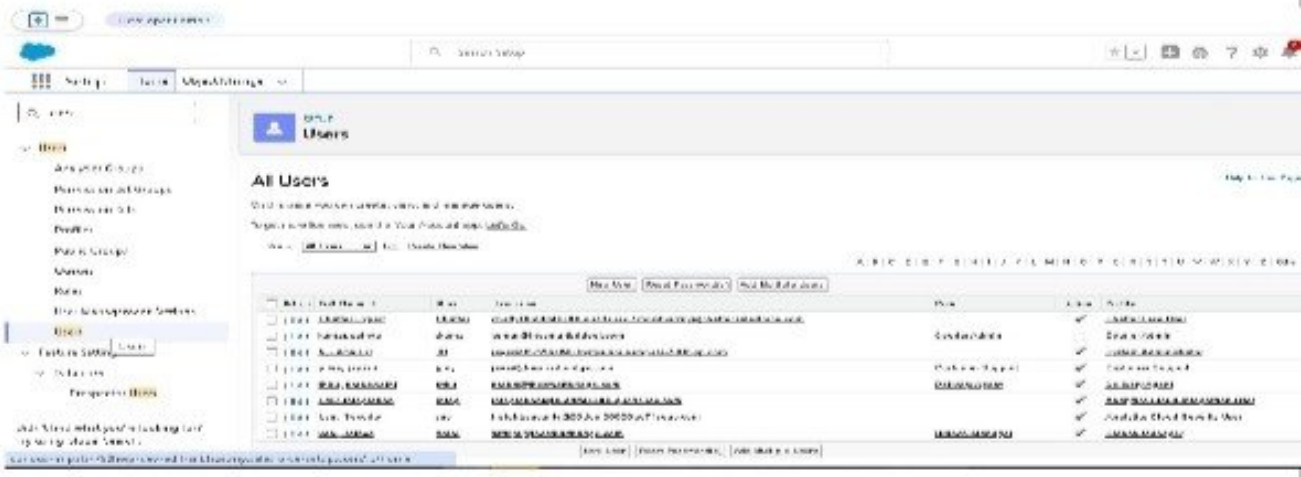
What is a user?

A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, So If you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

Activity 1: To Create a User

Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.
3. 

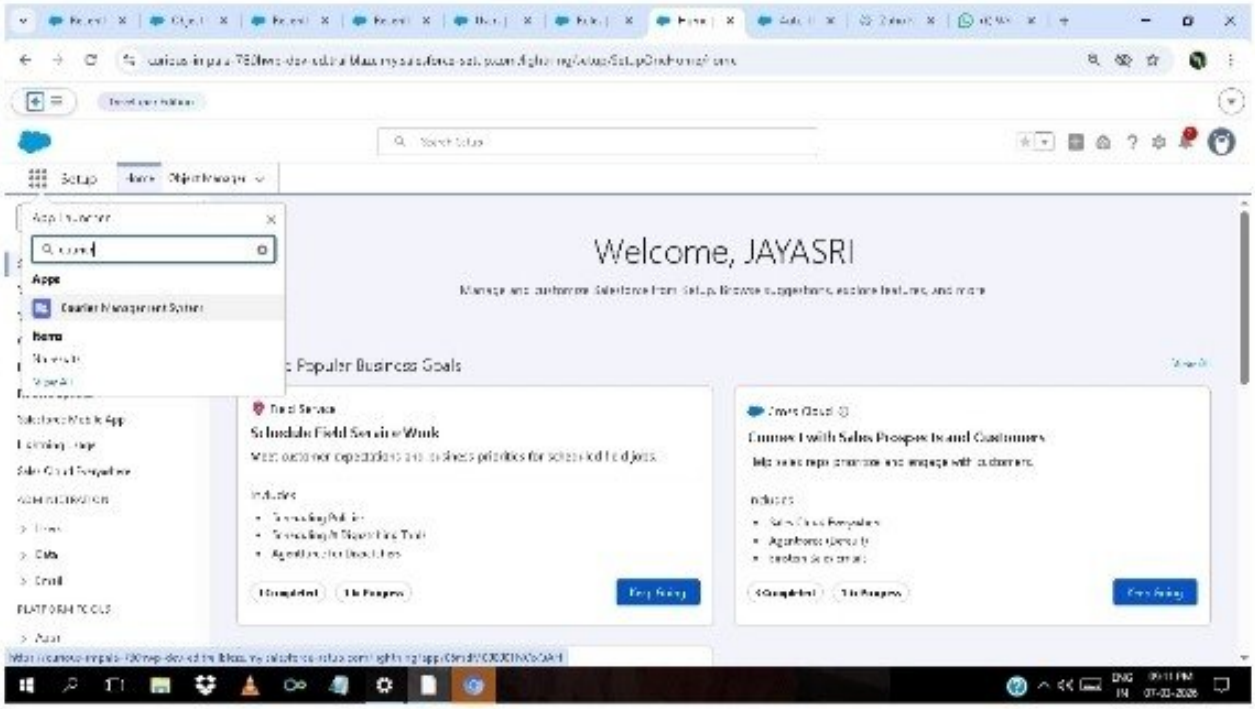
Milestone 15: Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

Step 1: Open Report Types

1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.



Step 2: Create Shipment Report Type

- Primary Object: Shipment_c
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 3: Create Customer Report Type

- Primary Object: Customer_c
- Label: Customer Summary Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 4: Create Agent Performance Report Type

- Primary Object: Delivery_Agent_c
- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Activity 2: Create Reports

Report 1: Shipment Status Summary Report

Step 1: Create the Report

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click

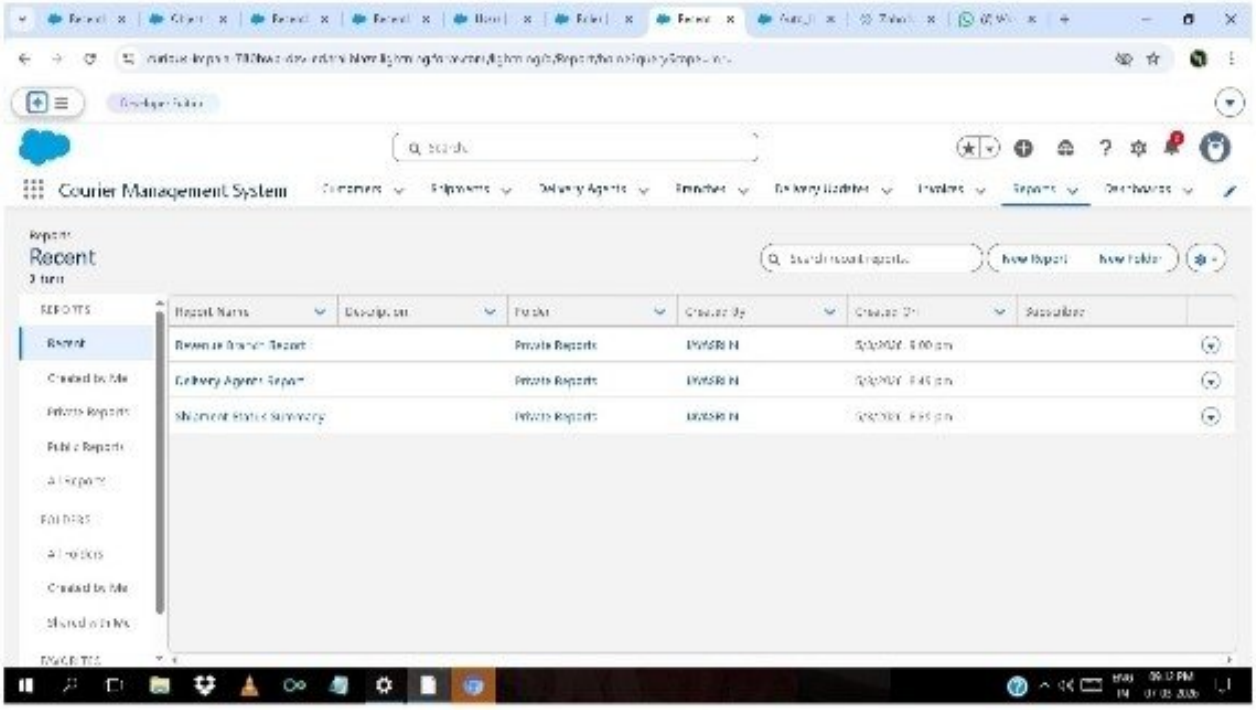
Activity 3: Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create



Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report Chart Type: Bar Chart
- Click Add

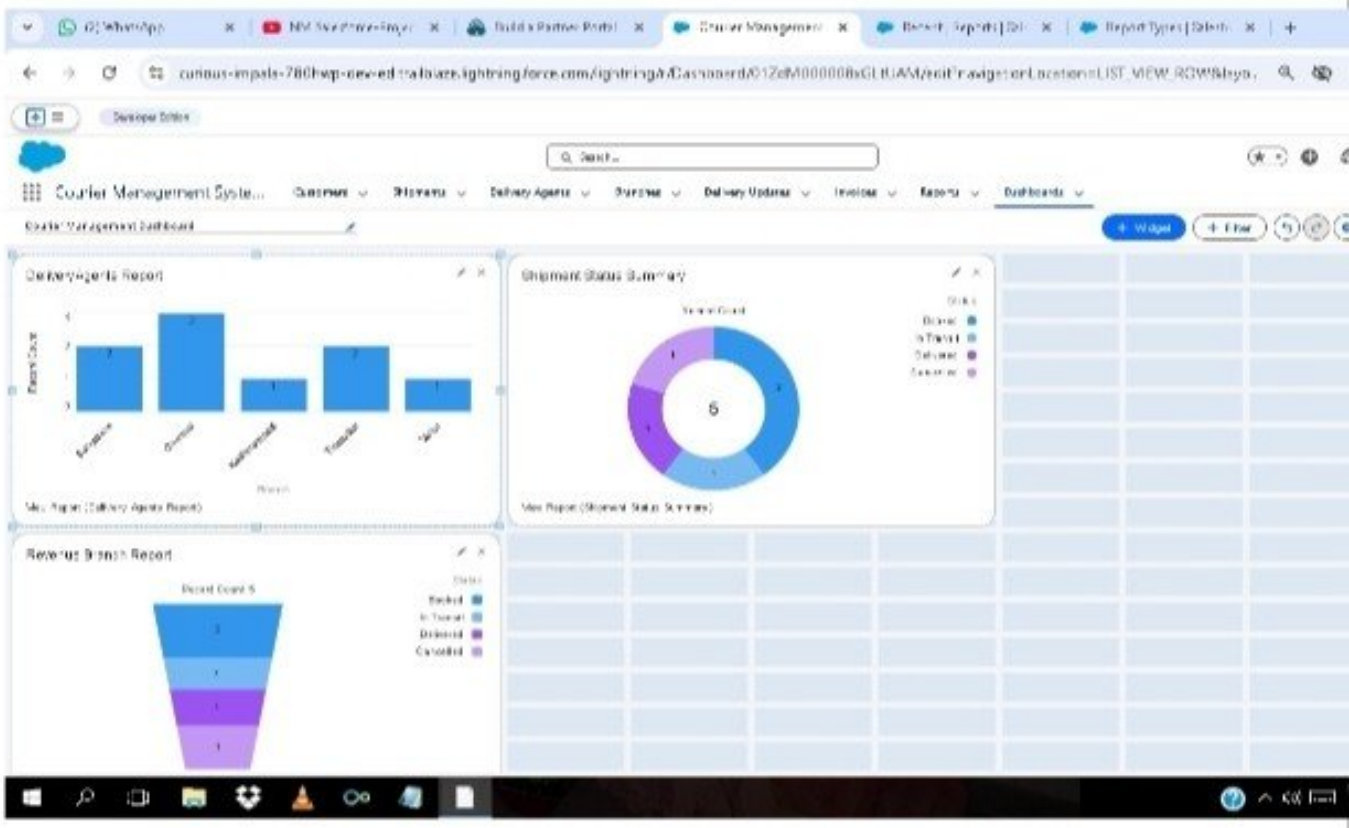
Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

Activity 5: Save & Activate Dashboard

1. Click Save
2. Click Done

Final Layout:



7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

Thank You